Escola Superior d'Enginyeries Industrial, Aeroespacial i Audiovisual de Terrassa

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1.0.1 The Eigenvalue problem

The free-vibration equation of motion of an undamped MDOF system, written in symbolic matrix form, is

$$\boldsymbol{M}\ddot{X} + \boldsymbol{K}X = \cancel{\boldsymbol{F}}\,0 \tag{1}$$

An undamped structure, with no external loads applied to unrestrained dof undergoes harmonic motion. $\overline{X}$ is the amplitude of the modal motion and ω is the cyclic frequency in radians per unit time.

$$X\left(t\right) = \overline{X}e^{i\omega t} \tag{2}$$

$$\ddot{X}\left(t\right) = -\overline{X}\omega^2 e^{i\omega t} \tag{3}$$

Susbtituting the harmonic solution equation in Equation 1,

$$-\boldsymbol{M}\overline{X}\omega^2 e^{i\omega t} + \boldsymbol{K}\overline{X}e^{i\omega t} = 0 \tag{4}$$

$$-\boldsymbol{M}\overline{X}\omega^2 \cancel{e^{i\omega t}} + \boldsymbol{K}\overline{X}\cancel{e^{i\omega t}} = 0 \tag{5}$$

perquè?

$$-\boldsymbol{M}\overline{X}\omega^2 + \boldsymbol{K}\overline{X} = 0 \tag{6}$$

We decide that $\lambda = \omega^2$

$$\lambda\boldsymbol{M}\overline{X} - \boldsymbol{K}\overline{X} = 0 \tag{7}$$

$$\boxed{\left[\boldsymbol{K} - \lambda\boldsymbol{M}\right]\overline{X} = 0} \tag{8}$$

For there to be a nontrivial solution of the previous equation it is necessary that,

$$det\left(\boldsymbol{K} - \lambda\boldsymbol{M}\right) = 0 \tag{9}$$

This is called the characteristic equation. When de determinant of the previous equation is expanded, there results a polynominal equation of degree N in λ, whose roots are the eigenvalues. For each eigenvalue results a natural frequency ω_i. Corresponding to each eigenvalue $\lambda_i$there will be an eigenvector, or natural mode ϕ_i.

Tipically, the eigenvalue problem is also written in the following form

$$\lambda\boldsymbol{M}^{-1}\boldsymbol{M}\overline{X} - \boldsymbol{M}^{-1}\boldsymbol{K}\overline{X} = 0 \tag{10}$$

$$\lambda\boldsymbol{I}\overline{X} - \boldsymbol{M}^{-1}\boldsymbol{K}\overline{X} = 0 \tag{11}$$

$$\boxed{\lambda\boldsymbol{I}\overline{X} = \left[\boldsymbol{M}^{-1}\boldsymbol{K}\right]\overline{X}} \tag{12}$$

In general, a structure has as many eigenmodes as unconstrained degrees of freedom. For this reason, eigenmodes extraction algorithms are designed in order to find the modes sequentially within a frequency range. MATLAB uses the function *eigs*[1] to extract a subset of eigenvalues and eigenvectors.

[1] https://es.mathworks.com/help/matlab/ref/eigs.html

The natural modes solution must satisfy all prescribed boundary conditions, that is, all displacement-type boundary conditions. We must discard all degrees of freedom which are constrained.

$$[\boldsymbol{K} - \lambda \boldsymbol{M}]\,\overline{X} = 0 \tag{13}$$

$$[\boldsymbol{K}_{NN} - \lambda \boldsymbol{M}_{NN}]\,\overline{X} = 0 \tag{14}$$

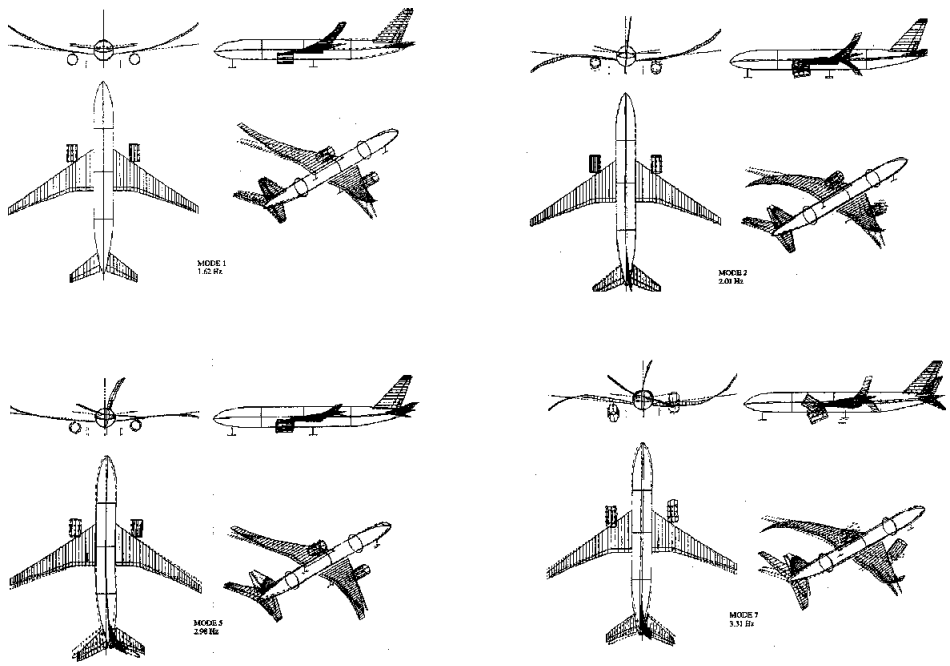

Figure 1: Aircraft natural modes

1.0.2 Direct frequency response

The equation of motion of an undamped MDOF system, written in symbolic matrix form, is

$$\boldsymbol{M}\ddot{X} + \boldsymbol{K}X = F \tag{15}$$

An undamped structure, with harmonic loads of frequency ω_i applied to unrestrained dof undergoes harmonic motion of frequency ω_i. $\overline{X}$ is the amplitude of the modal motion and ω is the cyclic frequency in radians per unit time.

Both excitation and the displacement of the unconstrained degrees of freedom can be represented with the harmonic solution equation. The frequency is not an unknown, it is a parameter of the excitation.

$$F = \overline{F}e^{i\omega t} \tag{16}$$

amplitud

sabem la massa, i decidim la w (frequencia d'exitació)

$$X = \overline{X}e^{i\omega t} \tag{17}$$

$$\ddot{X} = -\overline{X}\omega^2 e^{i\omega t} \tag{18}$$

We can substitute this equations into the dynamics equation

$$-\boldsymbol{M}\overline{X}\omega^2 e^{i\omega t} + \boldsymbol{K}\overline{X}e^{i\omega t} = \overline{F}e^{i\omega t} \tag{19}$$

$$-\boldsymbol{M}\overline{X}\omega^2 \cancel{e^{i\omega t}} + \boldsymbol{K}\overline{X}\cancel{e^{i\omega t}} = \overline{F}\cancel{e^{i\omega t}} \tag{20}$$

$$\underbrace{[-\boldsymbol{M}\omega^2 + \boldsymbol{K}]}_{\boldsymbol{Q}(\omega)}\overline{X} = \overline{F} \tag{21}$$

The matrix $\boldsymbol{Q}$ has to be computed for each freqyency of study. The solution of the problem is a vector of displacements amplitudes $\overline{X}$. The system of equations is then solved similarly to a static problem.

$$\boldsymbol{Q}\overline{X} = \overline{F} \tag{22}$$

$$\overline{X} = \boldsymbol{Q}^{-1}\overline{F} \tag{23}$$

The constrained degrees of freedom have to be taken into account by discarding them from the system of equations.

$$\overline{X}_N = \boldsymbol{Q}_{NN}^{-1}\overline{F}_N \tag{24}$$

If the forcing frequency is close to any one of the natural frequencies of the system, huge vibration amplitudes occur. This phenomenon is known as resonance. If the system is undamped, at these frequencies the vibration amplitude is theoretically infinite. At certain frequencies, the amplitude of vibration of some degrees of freedom drops to zero. This is called 'Anti-resonance,' and it has an important engineering application.

1.0.3 Modal frequency response

Modal frequency response analysis is an alternate approach to computing the frequency response of a structure. This method uses the mode shapes of the structure to reduce the size, uncouple the equations of motion (when modal or no damping is used), and make the numerical solution more efficient. Since the mode shapes are typically computed as part of the characterization of the structure, modal frequency response is a natural extension of a normal modes analysis. As a first step in the formulation, transform the variables from physical coordinates X to modal coordinates ξ. r of the grid points.

$$\overline{X} = \boldsymbol{\phi}\bar{\xi} \tag{25}$$

$$\boldsymbol{\phi}^T\overline{X} = \bar{\xi} \tag{26}$$

$$\overline{F} = \boldsymbol{\phi}\overline{F}^{\xi} \tag{27}$$

$$\boldsymbol{\phi}^T\overline{F} = \overline{F}^{\xi} \tag{28}$$

The modal shapes ϕ are used to transform the problem in terms of the behavior of the modes as opposed to the behavior. Eq. 25 represents an equality if all modes are used; however, because all modes are rarely used, the equation usually represents an approximation.

$$-\omega^2\boldsymbol{M}\overline{X} + \boldsymbol{K}\overline{X} = \overline{F} \tag{29}$$

$$-\omega^2\boldsymbol{M}\phi\bar{\xi} + \boldsymbol{K}\phi\bar{\xi} = \overline{F} \tag{30}$$

$$-\omega^2\phi^T\boldsymbol{M}\phi\bar{\xi} + \phi^T\boldsymbol{K}\phi\bar{\xi} = \underbrace{\phi^T\overline{F}}_{F^{\xi}} \tag{31}$$

$$\underbrace{\left(-\omega^2\phi^T\boldsymbol{M}\phi + \phi^T\boldsymbol{K}\phi\right)}_{\boldsymbol{Q}(\omega)}\bar{\xi} = F^{\xi} \tag{32}$$

$$\boldsymbol{Q}\bar{\xi} = F^{\xi} \tag{33}$$

The final step uses the orthogonality property of the mode shapes to formulate the equation of motion in terms of the generalized mass and stiffness matrices, which are diagonal matrices.[2] These diagonal matrices do not have the off-diagonal terms that couple the equations of motion. Therefore, in this form the modal equations of motion are uncoupled. In this uncoupled form, the equations of motion are written as a set of uncoupled single degree-of-freedom systems as

$$\left(-\omega^2 m_i^{\xi} + k_i^{\xi}\right)\xi_i = f_i^{\xi} \tag{34}$$

$$\xi_i = \frac{f_i^{\xi}}{-\omega^2 m_i^{\xi} + k_i^{\xi}} \tag{35}$$

However, we are not interested in amplitudes in modal coordinates but in cartesian coordinates.

$$\overline{X} = \boldsymbol{\phi}\bar{\xi} \tag{36}$$

[2] If M and Ka re symmetric and real (as is the case for all the common structural finite elements)

1.0.4 Damping implementation

For linear damping, the damping is either linearly proportional to the velocity $\dot{x}(t)$, the so-called viscous damping, or is linearly proportional to the elastic force $kx(t)$, the so-called structural damping. Hence the damping force can be defined as:

- Viscous damping $f_d = c\dot{x}$
- Structural damping $f_d = igkx$

At the resonance frequency of a harmonic vibration, both viscous damping and structural damping forces exhibit a positive phase shift of $\frac{\pi}{2} rad$ with respect to the elastic force. Structural damping is proportional to the stiffness. The phase shift of $\frac{\pi}{2}$ rad for structural damping is obtained by multiplying the proportional part of the elastic force with the complex number $\sqrt{-1}$.

The introduction of complex numbers in the mathematical model limits the application of the structural damping only for harmonic vibration.

1. Damping in Direct Frequency Response
   - Viscous elements of known $b\left[\frac{Ns}{mm}\right]$ such as a damper, they form the matrix B. This damping is usually called viscous damping and it is associated to the amplitude of the displacements.
   - In frequency response, overall structural damping G does not form a damping matrix. Instead, it forms the following complex stiffness matrix. The parameter G is obtained experimentally and is usually given in bibliography according to the characteristics of the structure. This damping is usually called hysteresis damping and it is associated to the amplitude of the displacements.

$$\boldsymbol{K}^D = (1 + iG)\,\boldsymbol{K} \tag{37}$$

2. Damping in Modal Frequency Response
   - If a damping matrix B exists, the orthogonality property of the modes does not, in general, diagonalize the generalized damping matrix. Also, if structural damping is used, the orthogonality property does not, in general, diagonalize the generalized stiffness matrix,

$$\boldsymbol{\phi^T B \phi} \neq diagonal \tag{38}$$

$$\boldsymbol{\phi}^T\left[(1 + iG)\,\boldsymbol{K}\right]\phi \neq diagonal \tag{39}$$

   - **Solution A**, Because the equations are not uncoupled, solve directly in modal coordinates. This is still efficient because the size of the matrix being solved is greatly reduced by the change of coordinates.
   - **Solution B**, if damping is applied to each mode separately, the uncoupled equations of motion can be maintained. When modal damping is used, each mode has damping $b_i = 2m_i\omega_i\zeta_i$, where ζ_i is the modal damping ratio and can be obtained experimentally. This is the standard way of proceeding in modal frequency response.

$$\xi_i = \frac{f_i}{-\omega^2 m_i + k_i + ib\omega} \tag{40}$$